

JAVA LAB RECORD

Student Performance Analysis System

1. Aim / Objective

To design and implement a student performance analysis system in Java using Object-Oriented Programming principles to calculate marks, percentage, grade, highest and lowest marks, and overall academic performance.

2. Problem Statement / Description

Develop a Java application for a college to analyze the academic performance of a student. The system stores five subject marks in an array and calculates total, average, highest, lowest, percentage, grade, pass/fail result, and a performance remark. It also formats the student's name using String operations such as trim(), toUpperCase(), and length(), and rounds the percentage to two decimal places using Math.round(). The program demonstrates class, object, constructor, methods, arrays, String operations, Math methods, and basic OOP.

3. Real-Time Application

College student performance analysis and academic report generation.

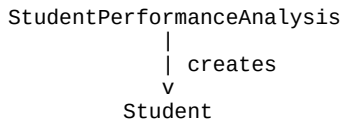
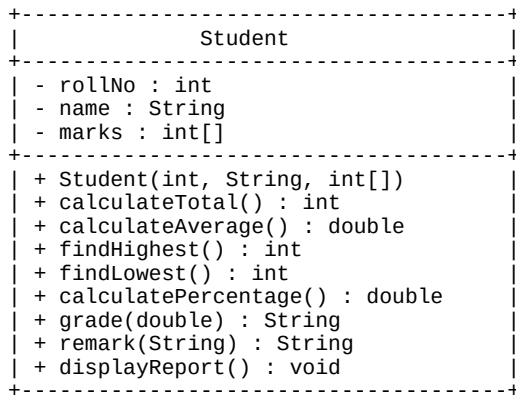
4. Algorithm

1. Start the program.
2. Create a Student class with roll number, name, and marks array.
3. Create a constructor to initialize student details.
4. Store marks of five subjects in an integer array.
5. Use a loop to calculate the total marks.
6. Calculate the average and percentage from the total marks.
7. Find the highest mark using Math.max() and the lowest mark using Math.min().
8. Round the percentage using $\text{Math.round}(\text{percentage} * 100.0) / 100.0$.
9. Process the name using trim(), toUpperCase(), and length().
10. Assign grade using the given percentage criteria.
11. If percentage is at least 50%, display PASS; otherwise display FAIL.
12. Display a performance remark based on the grade.
13. Create a Student object in main() and display the complete report.
14. Stop the program.

5. Grade Criteria

Percentage	Grade	Result
90–100	A+	Pass
80–89	A	Pass
70–79	B	Pass
60–69	C	Pass
50–59	D	Pass
Below 50	F	Fail

6. UML Class Diagram



7. Source Code

```
import java.util.*;

class Student {
    int rollNo;
    String name;
    int[] marks;

    Student(int rollNo, String name, int[] marks) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }

    int calculateTotal() {
        int total = 0;
        for (int mark : marks)
            total += mark;
        return total;
    }

    double calculateAverage() {
        return calculateTotal() / (double) marks.length;
    }

    int findHighest() {
        int high = marks[0];
        for (int mark : marks)
            high = Math.max(high, mark);
        return high;
    }

    int findLowest() {
        int low = marks[0];
        for (int mark : marks)
            low = Math.min(low, mark);
        return low;
    }

    double calculatePercentage() {
        return (calculateTotal() / (double) (marks.length * 100)) * 100;
    }

    String grade(double percentage) {
        if (percentage >= 90) return "A+";
        else if (percentage >= 80) return "A";
        else if (percentage >= 70) return "B";
        else if (percentage >= 60) return "C";
        else if (percentage >= 50) return "D";
        else return "F";
    }

    String remark(String grade) {
        if (grade.equals("A+")) return "Excellent Performance";
        if (grade.equals("A")) return "Very Good Performance";
        if (grade.equals("B")) return "Good Performance";
        if (grade.equals("C")) return "Satisfactory Performance";
    }
}
```

```

        if (grade.equals("D")) return "Needs Improvement";
        return "Fail - Needs Improvement";
    }

    void displayReport() {
        String formattedName = name.trim().toUpperCase();
        double percentage = calculatePercentage();
        double roundedPercentage = Math.round((percentage * 100.0) / 100.0);
        String gradeValue = grade(roundedPercentage);

        System.out.println("=====");
        System.out.println("          STUDENT PERFORMANCE ANALYSIS");
        System.out.println("=====");
        System.out.println("Roll Number : " + rollNo);
        System.out.println("Student Name: " + formattedName);
        System.out.println("Name Length : " + formattedName.length());
        System.out.println("Marks       : " + Arrays.toString(marks));
        System.out.println("Total Marks : " + calculateTotal() + "/" + (marks.length * 100));
        System.out.printf("Average       : %.2F%n", calculateAverage());
        System.out.println("Highest      : " + findHighest());
        System.out.println("Lowest       : " + findLowest());
        System.out.println("Percentage   : " + roundedPercentage + "%");
        System.out.println("Grade        : " + gradeValue);
        System.out.println("Result       : " + (percentage >= 50 ? "PASS" : "FAIL"));
        System.out.println("Remark       : " + remark(gradeValue));
        System.out.println("=====");
    }
}

public class StudentPerformanceAnalysis {
    public static void main(String[] args) {
        int[] marks = {92, 85, 78, 88, 95};

        Student student = new Student(101, " Biddika Rukmini ", marks);
        student.displayReport();
    }
}

```

8. Output Screenshot

Sample console output screenshot for the program:

```
=====
                        STUDENT PERFORMANCE ANALYSIS
=====
Roll Number : 101
Student Name: BIDDIKA RUKMINI
Name Length : 16
Marks       : [92, 85, 78, 88, 95]
Total Marks : 438/500
Average     : 87.60
Highest     : 95
Lowest      : 78
Percentage  : 87.6%
Grade       : A
Result      : PASS
Remark      : Very Good Performance
=====
```

9. Execution / Compilation Commands

```
Save as: StudentPerformanceAnalysis.java
Compile: javac StudentPerformanceAnalysis.java
Run:     java StudentPerformanceAnalysis
```

10. Result

The Student Performance Analysis System was successfully implemented in Java using OOP concepts, arrays, String operations, Math methods, and built-in packages. The program successfully calculates and displays total marks, average, highest and lowest marks, percentage, grade, result, and performance remark.